ADS-599 Capstone Project Proposal

Fill out this form and submit it by the end of Module 1 in Canvas.

(A) Team Formation:

1. Project Group #: Group 1

2. Team Leader/Representative: Franky Hernandez

3. Full Names of Team Members:

    1. Nikita Rogers
    2. Franky Hernandez
    3. Jun Sik Ryu

(B) Team Workflows:

1. Provide your team GitHub link here:
    - https://github.com/darth-franky/USD-ADS-Capstone
2. How many times have your team member(s) met during Module 1?
    - 5 times
3. What was the agreed upon method of communication? Are you using any teamwork project management software?
    - Communication via Slack. Currently using GitHub for teamwork project management.

(C) Capstone Project Dataset:

1. Title of your Capstone Project:
    - Forecasting California's Day-Ahead Electricity Demand: A Machine-Learning Decision-Support Tool for Peak-Load Planning
2. Name of Your Selected Dataset:
    - EIA-930 Hourly Electric Grid Monitor, California Independent System Operator (CISO) balancing authority
3. Description of Your Selected Dataset (data source, number of variables, size of dataset, etc.):
    - The dataset comes from the U.S. Energy Information Administration (EIA) Hourly Electric Grid Monitor (Form EIA-930). We will analyze hourly CISO records from 2019 through 2025. These seven complete years contain 61,368 nominal hourly timestamps before cleaning for missing, duplicate, or daylight-saving-time records. After filtering and pivoting the balance data, the primary analysis table will retain five source variables: hour-ending timestamp, actual demand, day-ahead demand forecast, net generation, and total interchange. The modeling table will add hour, weekday, month, holiday indicators, demand lags of 24, 48, and 168 hours, and rolling statistics. CISO is a

balancing-authority footprint and is not identical to statewide California demand. Sources: EIA-930 documentation and Hourly Electric Grid Monitor.

(D) Capstone Project Proposal:

The following proposal describes the work the team intends to complete. All results and deliverables are planned, not yet completed.

1. Purpose / The Problem

Electricity system operators must anticipate changes in demand to plan generation and maintain grid reliability. Demand can vary substantially by hour, weekday, season, and exceptional events, making day-ahead forecasting a meaningful data science problem. Our central research question is: How accurately can historical electricity demand and calendar-based variables predict electricity demand for each hour of the following day within the CISO balancing authority? In addition to forecasting hourly load, we will examine whether the models identify the timing and magnitude of daily peak demand. Our aim is to build a transparent, reproducible demonstration of a forecasting workflow rather than to replace operational forecasts.

2. Background

Short-term electricity load forecasting is an established application of statistical and machine-learning methods. Previous research has examined a range of approaches, from conventional models to more complex learning algorithms (Aguilar Madrid & Antonio, 2021; Ibrahim et al., 2022; Nti et al., 2020). The EIA-930 program provides historical hourly electricity-system measurements that can support a case study of the CISO region (U.S. Energy Information Administration, n.d.). A central methodological challenge is that day-ahead forecasts must use only information available when the forecast is issued. This constraint affects the use of lagged demand, rolling averages, and any potential weather predictors.

3. Your Motivation

We selected this topic because it combines a practical planning challenge with a manageable public dataset and an opportunity to communicate findings to nontechnical users. Electricity demand is relevant to system operations, resource planning, and the broader transition toward a changing energy mix. For our capstone, the project also provides a chance to demonstrate end-to-end data science skills: obtaining data, engineering time-series features, testing models, interpreting predictions, and deploying a usable application. Focusing on peak hours gives the work a clear decision-support use case beyond reporting an overall accuracy score.

4. Your Working Hypothesis

We hypothesize that models combining recent historical demand with calendar-based features will outperform a seasonal-naive forecasting baseline on held-out data. We also expect that a tree-based ensemble may capture nonlinear demand patterns better than regularized linear regression, although this will be tested rather than assumed. We expect forecast errors to be larger during unusual demand events and at certain daily peaks. Success will be judged using both overall hourly errors and the accuracy of predicted daily peak timing and magnitude.

5. Data Science Objectives

First, we will create a reproducible data-acquisition and preprocessing pipeline for the 2019–2025 CISO hourly dataset. Second, we will characterize hourly, daily, weekly, and seasonal demand patterns through exploratory analysis. Third, we will engineer calendar and historical-demand features while preventing future-data leakage. Fourth, we will compare a seasonal-naive benchmark, a regularized linear regression model, and at least one tree-

based ensemble such as Random Forest, XGBoost, or LightGBM. Fifth, we will assess general forecast accuracy and peak-demand prediction performance on a held-out period. Finally, we will present the findings in a Streamlit dashboard and publish documented, reproducible project code.

6. Planned Methodology

We will retrieve and filter EIA-930 records to the CISO balancing authority and inspect completeness, duplicates, outliers, and time-zone or daylight-saving-time irregularities. After establishing a consistent hourly time index, we will perform exploratory analysis and construct calendar variables, holiday flags, lagged demand, and rolling summaries. We will define a specific daily forecast issuance time and ensure that every predictor is known at that cutoff; for example, a 24-hour lag is not automatically available for all hours of a next-day forecast. We will use chronological training and validation splits, preferably with rolling-origin evaluation for model selection, and reserve 2025 as a final holdout test period. Models will be compared using mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE), with appropriate handling of any zero or problematic denominators. We will additionally evaluate daily peak-load magnitude error and the difference between predicted and actual peak hours. We will examine feature importance using a suitable interpretation method. Weather data and deep learning are stretch goals only if the core workflow is complete and any added inputs are available at forecast issuance.

7. Real-World Impact

The intended outcome is an accessible example of how predictive analytics can help users understand day-ahead load variation and recognize potential high-demand periods. The project may be useful for students, analysts, and stakeholders interested in load forecasting, but its results should not be treated as operational recommendations. By comparing simple and more complex models and documenting their errors, the work will demonstrate the tradeoffs between predictive performance, interpretability, and reliability. We will explicitly discuss limitations related to data revisions, unusual events such as COVID-era demand shifts, the geographic scope of CISO, and missing predictors such as weather.

8. Project Artifact

The final artifact will be an interactive Streamlit application tentatively titled “California Peak Demand Planner.” Users will be able to explore historical hourly demand; review daily, weekly, and seasonal patterns; select a date within the test period; compare actual versus predicted demand for each hour; identify predicted and actual peak-demand hours; compare model-performance metrics; review influential prediction variables; and read explanations of the data, methods, and limitations. The initial application will present historical backtesting results, not claim to generate live operational forecasts. A companion GitHub repository will include data-acquisition instructions, preprocessing and exploratory-analysis code, model-training workflows, saved evaluation outputs, and application deployment instructions.

References

Aguilar Madrid, E., & Antonio, N. (2021). Short-term electricity load forecasting with machine learning. *Information, 12*(2), Article 50. https://www.mdpi.com/2078-2489/12/2/50

Ibrahim, B., Rabelo, L., Gutierrez-Franco, E., & Clavijo-Buritica, N. (2022). Machine learning for short-term load forecasting in smart grids. *Energies, 15*(21), Article 8079. https://www.mdpi.com/1996-1073/15/21/8079

Nti, I. K., Teimeh, M., Nyarko-Boateng, O., & Adekoya, A. F. (2020). Electricity load forecasting: A systematic review. *Journal of Electrical Systems and Information Technology, 7*, Article 13. https://link.springer.com/article/10.1186/s43067-020-00021-8

U.S. Energy Information Administration. (n.d.). *About the EIA-930 data*. Retrieved October 26, 2026, from https://www.eia.gov/electricity/930-content/about.html

(E) Additional Comments / Roadblocks:

LLM Use Acknowledgment
OpenAI ChatGPT was used during project ideation to compare potential data sources. The team remains responsible for verifying all dataset details and references and for the final content submitted.

I assert that this proposal is the result of my own work, and all applicable sources have been properly cited and acknowledged. I understand that my capstone project will be reviewed by the Program Director as part of the approval process and successful completion of this project requires approval from the Program Director and/or Course Instructor. In addition, I understand that I am strongly encouraged to submit my finished project for publication.

Student Signature	**Date**
Jun Sik Ryu	**10/26/2026**
Student Signature	**Date**
Francisco Hernandez	**10/26/2026**
Student Signature (if applicable)	**Date**
Nikita Rogers	**10/26/2026**